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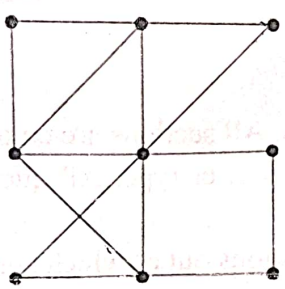
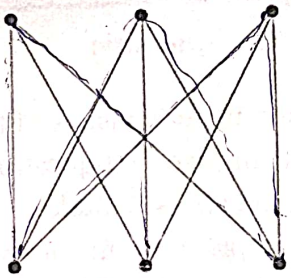
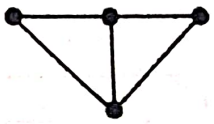
Course: B. Tech Computer Engineering / B. Tech Computer Science and Engineering (Artificial Intelligence & Machine LEARNING)
Subject: Discrete Mathematics
Subject Code: ETCS-201
Semester: III

Time: 03 Hours **Max Marks: 70**

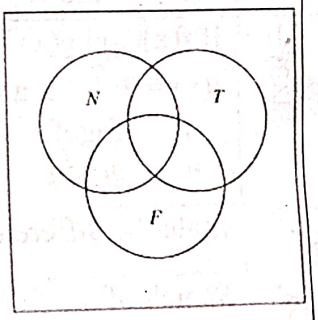
Instructions to the Students:

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 2 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

Q.No.		SECTION –A (SHORT ANSWER TYPE QUESTIONS)	Marks
1.	a.	Consider the relation R = {(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)} on A = {1, 2, 3, 4}. i. Draw its directed graph. ii. Find R^2 = RoR.	(2)
	b.	If f(x) and g(x) be the function from these the set of integers defined as $f(x) = 2x^2 + 2$ and $g(x) = 3x + 3$ then determine the following: i. fogof, ii. gofog.	(2)
	c.	State the difference between permutations and combinations with examples.	(2)
	d.	Which of the following set(s) follow the Well-Ordering Principle: i. Set of Natural numbers, ii. Set of Real numbers, iii. Set of Even numbers, iv. Set of Integers.	(2)
	e.	Find the truth tables for: i. $p \vee \neg q$; ii. $\neg p \wedge \neg q$.	(2)
	f.	Let p(x) denote the sentence "x + 2 > 5". State whether or not p(x) is a propositional function on each of the following sets: i. M = {−1,−2,−3, . . .}; ii. C, the set of complex numbers.	(2)

	g	Consider the set Q of rational numbers, and let $*$ be the operation on Q defined by $a * b = a + b - ab$. Find: i. $3 * 4$; ii. $2 * (-5)$; iii. $7 * (1/2)$; iv. the identity element for $*$?	(2)
	h.	Define abelian group and monoid with example?	(2)
	i.	Considering the graphs given below, answer the following questions: Which of them is traversable, i.e., have Euler paths? If no, why? Which of them is Eulerian, i.e., have an Euler circuit? If no, why?  a)  b)	(2)
	j.	Draw all spanning trees of the following graph G: 	(2)

SECTION –B (LONG ANSWER TYPE QUESTIONS)

2.	a)	Let f: R → R be defined by $f(x) = 2x - 3$. Now f is one-to-one and onto; hence f has an inverse function f^{-1}. Find a formula for f^{-1}.	(5)
	b)	In a survey of 120 people, it was found that: 65 read Newsweek magazine, 45 read Time, 20 read both Newsweek and Time, 42 read Fortune, 25 read both Newsweek and Fortune, 15 read both Time and Fortune, 8 read all three magazines. i) Find the number of people who read at least one of the three magazines. ii) Fill in the correct number of people in each of the eight regions of the Venn diagram below where N, T, and F denote the set of people who read Newsweek, Time, and Fortune, respectively. iii) Find the number of people who read exactly one magazine. 	(5)
3.	a)	Prove that the set I of all real numbers between 0 and 1 inclusive is uncountable.	(5)
	b)	In a certain school, French (F), Spanish (S), and German (G) are the only foreign languages taught. Among 80 students: (i) 20 study F, 25 study S, 15 study G. (ii) 8 study F and S, 6 study S and G, 5 study F and G. (iii) 2 study all three languages. Find the number of the 80 students who are studying:	(5)

	1) none of the languages; 2) only French; 3) only one language; 4) only Spanish and German; 5) exactly two of the languages.	
4.	a) State and prove Binomial Theorem. b) A class has 10 male students and 8 female students. Find the number of ways the class can elect: (i) a class representative; (ii) 2 class representatives, one male and one female; (iii) a class president and vice president.	(5) (5)
5.	a) Consider the third-order homogeneous recurrence relation: $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$ (i) Find the general solution. (ii) Find the solution with initial conditions as $a_0 = 3$, $a_1 = 4$, $a_2 = 12$. b) Prove the following argument is valid: $p \rightarrow \neg q$, $r \rightarrow q$, $r \vdash \neg p$.	(5) (5)
6.	a) Determine the validity of the following argument: If 7 is less than 4, then 7 is not a prime number. 7 is not less than 4. --- 7 is a prime number. b) Let A = {1, 2, . . . , 9, 10}. Consider each of the following sentences: If it is a statement, then determine its truth value. If it is a propositional function, determine its truth set. i. $(\forall x \in A)(\exists y \in A)(x + y < 14)$ ii. $(\forall y \in A)(x + y < 14)$ iii. $(\forall x \in A)(\forall y \in A)(x + y < 14)$ iv. $(\exists y \in A)(x + y < 14)$	(5) (5)
7.	a) Consider the group G = {1, 2, 3, 4, 5, 6} under multiplication modulo 7. i. Find the multiplication table of G. ii. Find $2^{-1}, 3^{-1}, 5^{-1}, 6^{-1}$. iii. Find the orders and subgroups generated by 2, 3, 5 and 6. iv. Is G cyclic?	(10)
8.	a) Prove that (G , *) is a cyclic group where $G=\{1,w,w^2\}$ with their composition table. b) Prove the theorem that the complete graph K_n with $n \geq 3$ vertices has $H = (n-1)! / 2$ Hamiltonian circuits.	(5) (5)
9.	a) State and prove five color theorem of graph theory by induction method? b) Let G be a graph with more than one vertex. Prove the following are equivalent. i. G is a tree. ii. Each pair of vertices is connected by exactly one simple path. iii. G is connected; but G − e is disconnected for any edge e of G. iv. G is cycle-free, but if any edge is added to G then the resulting graph has exactly one cycle.	(3) (7)

===END OF PAPER===